AC 547 Proposal

Are public companies with riskier profiles more likely to be audited by firms that perform poorly in PCAOB inspections?

We measure an audit firm's inspection performance by its Part I.A deficiency rate, which is the share of inspected audits where inspectors found the firm had not obtained sufficient evidence to support its opinion. We will test whether firms with higher deficiency rates concentrate in particular industries, among smaller clients, or among clients with signs of financial distress (losses, high leverage, weak liquidity).

Why is it important: Audit committees hire and oversee the external auditor, but they rarely compare firms on public inspection results. If the riskiest companies are paired with the weakest auditors, the companies that most need a strong audit are getting a weaker one. That raises the chance that misstatements go undetected. The findings also matter to investors judging how reliable reported earnings are, and to the PCAOB in targeting inspections.

How will we address it: We will build a client-level dataset in Python from three public sources:

- PCAOB inspection datasets (CSV, 2018 onward): For each firm and inspection year, the number of audits reviewed and the number with Part I.A deficiencies. Dividing the second by the first gives the deficiency rate.
- PCAOB AuditorSearch (Form AP): The audit firm (by PCAOB firm ID) for each public company (by CIK) and fiscal year.
- SEC Financial Statement Data Sets: Client industry (SIC code), size (total assets, revenue) and risk measures (net loss, leverage, current ratio).

We will then join and clean the data, then create summary statistics and visualizations to analyze the data and present our findings.

Revised Proposal

I revised our proposal to make the research design stronger and to address problems a reviewer would likely raise. The main changes:

1. **Added a limitations section.** The PCAOB mostly picks up the riskiest audits to inspect, so firms with risky clients may show higher deficiency rates even if their work is no worse. Auditors also choose clients, not just the other way around. Because of both, we'll present our results as associations, not cause and effect.
2. **Made the deficiency rate more reliable.** Small firms have only a few audits inspected, so one bad audit can produce an extreme rate. We'll pool inspection years (total deficient audits ÷ total inspected audits) and require a minimum number of inspected audits per firm.
3. **Fixed a timing issue.** We'll match each client only to inspection results that were public before its audit report date, so we aren't using information that wasn't available yet.
4. **Added comparisons by firm tier.** We'll compare clients within Big 4, other annually inspected firms, and triennially inspected firms, to reduce the effect of firm size.
5. Added a regression so we can test the relationship, not just describe it."

Are public companies with riskier profiles more likely to be audited by firms that perform poorly in PCAOB inspections?

We measure an audit firm's inspection performance by its Part I.A deficiency rate. This is the share of inspected audits in which inspectors found the firm had not obtained sufficient appropriate evidence to support its opinion. We will test whether firms with higher deficiency rates are concentrated in particular industries, among smaller clients, or among clients showing signs of financial distress (losses, high leverage, weak liquidity).

Why It Is Important

Audit committees hire and oversee the external auditor, but they rarely compare audit firms using public inspection results. If the riskiest companies are paired with audit firms that have higher inspection deficiency rates, the companies that most need a strong audit may receive audits that raise greater concerns about quality. That could increase the chance that misstatements go undetected. The findings also matter to investors judging how reliable reported earnings are, and to the PCAOB in deciding where to focus inspections.

How We Will Address It

We will build a client-level dataset in Python from three public sources:

- PCAOB inspection datasets (2018 onward): For each firm and inspection year, the number of audits reviewed and the number with Part I.A deficiencies. Dividing the second by the first gives the deficiency rate.
- PCAOB AuditorSearch (Form AP): The audit firm (by PCAOB firm ID) for each public company (by CIK) and fiscal year.
- SEC Financial Statement Data Sets: Client industry (SIC code), size (total assets, revenue), and risk measures (net loss, leverage, current ratio).
- We will also estimate a multiple regression of the auditor's deficiency rate on client risk measures (net loss, leverage, current ratio, and size), controlling for industry and audit firm tier, to test whether the relationships hold after accounting for these factors.

To make the deficiency rate a more reliable measure, we will take four steps:

1. Pool inspection years. Smaller firms are inspected about every three years, with only a few audits reviewed each time. We will combine inspection years and calculate each firm's rate as total deficient audits divided by total inspected audits, rather than averaging yearly percentages. We will also require a minimum number of inspected audits per firm, so that one deficient audit does not produce an extreme rate.
2. Use only previously published results. We will match each client to its auditor's deficiency rate using only inspection reports that were publicly available before the relevant audit report date. Because inspection reports are published a year or more after the audits they cover, this may reduce the number of client-years in the earliest part of the sample.
3. Group firms by tier. We will compare clients within firm tiers (Big 4, other annually inspected firms, and triennially inspected firms). This helps reduce the influence of differences in audit firm size.
4. Treat financial companies separately. Leverage and the current ratio do not mean the same thing for banks and insurers (SIC 6000–6999), so we will exclude them from the risk analysis or analyze them separately.

We will then join and clean the data, limit the financial data to annual 10-K figures, and standardize items that companies tag inconsistently, such as revenue. Finally, we will produce summary statistics and visualizations to present our findings.

Limitations

The PCAOB uses a risk-based approach to choose which audits to inspect. A firm with many risky clients will therefore have more risky audits reviewed, which may raise its deficiency rate even if its audit quality is no worse. In addition, auditor-client matching runs both ways: risky companies may choose firms with higher deficiency rates, and other firms may decline or resign from risky clients. For these reasons, we will present our results as associations between client risk and auditor inspection performance, not as evidence of cause and effect.

Sources:

1. PCAOB inspection data (deficiency rates)

The data is split by firm type across three pages. You'll likely need all three to cover the whole sample, including the triennially inspected firms.

- Inspection Data for U.S. Global Network Firms: Big 4, BDO, and Grant Thornton
- Inspection Data for Non-Affiliate Annual Firms: other firms inspected every year
- Non-Affiliated Firms Inspection Data: smaller firms, mostly inspected every three years
- PCAOB announcement of the datasets: what the datasets contain
- Firm Inspection Reports: the individual reports, if you need publication dates

2. PCAOB AuditorSearch (Form AP: who audits whom)

- AuditorSearch: the page offers a bulk download of all Form AP filings, so you don't need to search company by company.

3. SEC Financial Statement Data Sets (client size, industry, risk)

- Financial Statement Data Sets: one zip file per quarter
- Financial Statement Data Sets Archive: older quarters